

Chat Link: <https://share.gemini.google/gcPUMwskOrS7>

```
global_data = {  
  "attribute_1": "total_cost",  
  "attribute_2": "material_continuity",  
  "attribute_3": "electrothermal_performance",  
  "attribute_4": "mechanical_design",  
  "attribute_5": "manufacturability",  
  "direction_attribute_1": "negative",  
  "direction_attribute_2": "positive",  
  "direction_attribute_3": "positive",  
  "direction_attribute_4": "positive",  
  "direction_attribute_5": "positive",  
  "attribute_1_min": 0.30,  
  "attribute_1_max": 1.00,  
  "attribute_2_min": 0.10,  
  "attribute_2_max": 1.00,  
  "attribute_3_min": 0.45,  
  "attribute_3_max": 1.00,  
  "attribute_4_min": 0.35,  
  "attribute_4_max": 1.00,  
  "attribute_5_min": 0.50,  
  "attribute_5_max": 1.00  
}
```

```
regional_data = [  
  {  
    "region": "United States",  
    "copper_price_per_kg": 9.50,  
    "aluminum_price_per_kg": 2.80,  
    "copper_product_market_share": 0.75,  
    "aluminum_product_market_share": 0.25,  
    "weight_attribute_1": 0.20,  
    "weight_attribute_2": 0.20,  
    "weight_attribute_3": 0.25,  
    "weight_attribute_4": 0.20,  
    "weight_attribute_5": 0.15  
  },  
  {  
    "region": "China",  
    "copper_price_per_kg": 9.80,  
    "aluminum_price_per_kg": 2.15,  
    "copper_product_market_share": 0.60,  
    "aluminum_product_market_share": 0.40,  
  }  
]
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    "weight_attribute_1": 0.30,
    "weight_attribute_2": 0.15,
    "weight_attribute_3": 0.20,
    "weight_attribute_4": 0.15,
    "weight_attribute_5": 0.20
  },
  {
    "region": "India",
    "copper_price_per_kg": 10.10,
    "aluminum_price_per_kg": 2.55,
    "copper_product_market_share": 0.80,
    "aluminum_product_market_share": 0.20,
    "weight_attribute_1": 0.40,
    "weight_attribute_2": 0.10,
    "weight_attribute_3": 0.15,
    "weight_attribute_4": 0.15,
    "weight_attribute_5": 0.20
  },
  {
    "region": "Europe",
    "copper_price_per_kg": 9.55,
    "aluminum_price_per_kg": 2.65,
    "copper_product_market_share": 0.70,
    "aluminum_product_market_share": 0.30,
    "weight_attribute_1": 0.15,
    "weight_attribute_2": 0.25,
    "weight_attribute_3": 0.25,
    "weight_attribute_4": 0.20,
    "weight_attribute_5": 0.15
  }
]

```

```

product_data = [
  {
    "region": "United States",
    "dominant material": "cu",
    "product": "EV Busbar",
    "copper_kg": 1.45,
    "aluminum_kg": 0.00,
    "total_mass_kg": 1.45,
    "non_material_cost_per_unit": 7.50,
    "attribute_2_value": 0.80,
    "attribute_3_value": 0.90,
    "attribute_4_value": 0.75,

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"attribute_5_value": 0.65
},
{
  "region": "United States",
  "dominant material": "al",
  "product": "EV Busbar",
  "copper_kg": 0.00,
  "aluminum_kg": 0.75,
  "total_mass_kg": 0.75,
  "non_material_cost_per_unit": 8.10,
  "attribute_2_value": 0.20,
  "attribute_3_value": 0.70,
  "attribute_4_value": 0.60,
  "attribute_5_value": 0.55
},
{
  "region": "China",
  "dominant material": "cu",
  "product": "EV Busbar",
  "copper_kg": 0.95,
  "aluminum_kg": 0.00,
  "total_mass_kg": 0.95,
  "non_material_cost_per_unit": 3.80,
  "attribute_2_value": 0.55,
  "attribute_3_value": 0.70,
  "attribute_4_value": 0.65,
  "attribute_5_value": 0.90
},
{
  "region": "China",
  "dominant material": "al",
  "product": "EV Busbar",
  "copper_kg": 0.00,
  "aluminum_kg": 0.49,
  "total_mass_kg": 0.49,
  "non_material_cost_per_unit": 4.10,
  "attribute_2_value": 0.45,
  "attribute_3_value": 0.55,
  "attribute_4_value": 0.55,
  "attribute_5_value": 0.80
},
{
  "region": "India",
  "dominant material": "cu",
```

```
"product": "EV Busbar",
"copper_kg": 0.55,
"aluminum_kg": 0.00,
"total_mass_kg": 0.55,
"non_material_cost_per_unit": 2.90,
"attribute_2_value": 0.45,
"attribute_3_value": 0.60,
"attribute_4_value": 0.40,
"attribute_5_value": 0.80
},
{
  "region": "India",
  "dominant material": "al",
  "product": "EV Busbar",
  "copper_kg": 0.00,
  "aluminum_kg": 0.29,
  "total_mass_kg": 0.29,
  "non_material_cost_per_unit": 3.20,
  "attribute_2_value": 0.55,
  "attribute_3_value": 0.45,
  "attribute_4_value": 0.35,
  "attribute_5_value": 0.70
},
{
  "region": "Europe",
  "dominant material": "cu",
  "product": "EV Busbar",
  "copper_kg": 1.15,
  "aluminum_kg": 0.00,
  "total_mass_kg": 1.15,
  "non_material_cost_per_unit": 8.20,
  "attribute_2_value": 0.90,
  "attribute_3_value": 0.85,
  "attribute_4_value": 0.85,
  "attribute_5_value": 0.60
},
{
  "region": "Europe",
  "dominant material": "al",
  "product": "EV Busbar",
  "copper_kg": 0.00,
  "aluminum_kg": 0.60,
  "total_mass_kg": 0.60,
  "non_material_cost_per_unit": 9.00,
```

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    "attribute_2_value": 0.10,  
    "attribute_3_value": 0.65,  
    "attribute_4_value": 0.70,  
    "attribute_5_value": 0.50  
  }  
]
```